

100 Days of Azure DevOps

Day 42 Handout — Every Portal click is still an ARM call. JSON is broccoli: nutritious, rarely anyone's favorite

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — template anatomy (load-bearing boredom)

Key	Role	Lab value
\$schema	JSON schema for validation	2019-04-01/deploymentTemplate.json#
contentVersion	Your version, not apiVersion	10.0.0.1
parameters / variables	Dials and locals (deep dive Day 43)	Can be empty today
resources[]	type + apiVersion + name + properties	Microsoft.Storage/storageAccounts@2023-01-01
outputs	Handshake to the next stack	Optional today; required tomorrow
mode	Incremental (default) vs Complete	Complete can delete extras. Do not use Complete on a busy RG

Architecture B — how this relates to Bicep and Terraform

Tool	Talks to	You still need ARM literacy because
ARM JSON	ARM deployments API directly	Error text and export-template dumps are JSON
Bicep	Compiles to ARM, then the same API	Failed deploy details show ARM resource IDs / apiVersion
Terraform azurearm	ARM via the provider	The resource type names are the ARM types in snake_case
Portal export	A start that includes defaults you did not choose	Read it; do not worship it

Architecture C — pitfalls that deploy 2016 into 2026

Pitfall	What you see	Fix
Stale apiVersion	Property not allowed / missing SKU fields	Pin a current apiVersion from Learn, not a 2016 gist
Hardcoded storage name	AlreadyExists globally	uniqueString(resourceGroup().id) concat
Complete mode on shared RG	Unrelated resources deleted	Incremental. Separate rg-day42-lab
Leaving the account	LRS storage still meters	az group delete after the proof

One-liner to remember

ARM JSON is the broccoli of Azure — nutritious, rarely anyone's favorite. Eat a forkful so Bicep and Terraform errors make sense.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription. RG rg-day42-lab. Delete tonight. This is literacy even if Terraform is your primary.

- 1 az group create -n rg-day42-lab -l centralindia on the personal subscription.
- 2 Save infra/storage.json (starter). Confirm type Microsoft.Storage/storageAccounts and a real apiVersion.
- 3 az deployment group create -g rg-day42-lab -n stor -f infra/storage.json. Wait for Succeeded.
- 4 az storage account list -g rg-day42-lab -o table. Note the unique name ARM computed.
- 5 Open the deployment in Portal → RG → Deployments → stor. That history is ARM's receipt (no Terraform state here).
- 6 az group delete -n rg-day42-lab --yes --no-wait. Broccoli does not need to live in Cost Management.

Tiny storage ARM template + deploy

```
# infra/storage.json – save as JSON, then:
# az deployment group create -g rg-day42-lab -n stor -f infra/storage.json
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "10.0.0.1",
  "resources": [
    {
      "type": "Microsoft.Storage/storageAccounts",
      "apiVersion": "2023-01-01",
      "name": "[concat('st', uniqueString(resourceGroup().id))]",
      "location": "[resourceGroup().location]",
      "sku": { "name": "Standard_LRS" },
      "kind": "StorageV2",
      "properties": { "allowBlobPublicAccess": false }
    }
  ]
}
```

Done checklist

- Can name \$schema, contentVersion, resources, apiVersion
- One successful az deployment group create
- Know Incremental vs Complete well enough to fear Complete
- rg-day42-lab deleted

Tomorrow — Day 43

ARM parameters and outputs — dials in a .parameters.json, handshake via outputs, not hardcoded landfill names.